

# Question Paper

Exam Date & Time: 20-Nov-2025 (02:00 PM - 05:00 PM)



## MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU  
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Natural Language Processing [NPT\_4127]

Marks: 50

Duration: 180 mins.

### PART-A

Answer all the questions.

- 1) Analyze how regular expressions can be applied to solve real-world text processing tasks such as **email validation (user@example.com) and extracting dates from documents (DD-MM-YYYY)**. (4)
- A) Course Outcome: "CO1" / Cognitive Level: "Analysis"
- B) Identify the two interpretations of the structurally ambiguous sentence "**The police chased the man with a telescope.**" Describe the alternative syntactic structures and explain which meaning is more plausible based on context. (4)
- S → NP + VP  
NP → Det + N  
VP → V + NP + (PP)  
NP → NP + PP  
PP → P + NP  
NP → Det + N
- Course Outcome: "CO1" / Cognitive Level: "Application"
- C) **Explain** the concepts of Speech Processing and Language Processing in NLP by **describing** how they **differ** and how they **work together** to enable human-computer interaction. (2)
- Course Outcome: "CO1" / Cognitive Level: "Application"
- 2) Evaluate the efficiency and accuracy of using the *minimum edit distance algorithm* to transform intention into execution, given the operation costs (insert = 3, delete = 3, substitute = 5). (4)
- A) Course Outcome: "CO2" / Cognitive Level: "Evaluate"
- B) Evaluate the effectiveness of using regular expressions to detect email addresses (e.g., name@example.com) in large text corpora. *Discuss challenges such as handling subdomains or invalid formats, and suggest ways to improve accuracy.* (4)
- Course Outcome: "CO2" / Cognitive Level: "Evaluate"
- C) Explain the closure properties of FSTs ensure that complex linguistic transformations can still be represented using finite-state models. (2)
- Course Outcome: "CO2" / Cognitive Level: "Understand"
- 3) Evaluate the efficiency of the edit distance algorithm when transforming the word '*communication*' into '*computation*', given that the cost of insertion and deletion is 2, and substitution cost is 4. Justify (4)

- A) whether these cost choices produce an optimal alignment.

Course Outcome: "CO3" / Cognitive Level: "Evaluate"

- B) (4)

<s> I love learning new things </s>  
<s> Learning new things makes me happy </s>  
<s> She enjoys reading stories at night </s>  
<s> Stories she reads are full of adventure </s>  
<s> We do not like noisy places </s>  
<s> They like to travel during holidays </s>  
<s> Traveling helps them relax and explore </s>  
<s> The sun rises in the east </s>  
<s> The east is where the sun rises </s>  
<s> He plays football with his friends </s>  
<s> His friends enjoy playing football too </s>  
<s> I prefer tea but she likes coffee </s>  
<s> Coffee keeps her awake during work </s>  
<s> We do not watch television often </s>  
<s> Watching movies together is our weekend plan </s>

Apply the bigram probability estimation method to determine the probability of the test sentence "She like to travel" and also normalize the probability value with  $10^{-9}$

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Apply speech and discourse analysis techniques to the utterance 'I uh was plan- planning to meet her today. Well, we we um should start the meeting now.' (2)  
Identify and classify the fillers and broken-off words, and illustrate how recognizing such disfluencies can be used to improve the performance of an automatic speech recognition or dialogue system."

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) Examine the three methods of Part-of-Speech (POS) tagging by analysing how each method identifies word categories in a sentence. Compare their working principles and illustrate with suitable examples. (4)
- A)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Given the following sentences: (4)
- Predicted: - The (DT) quick (RB) brown (JJ) fox (VB) jumps (VBZ) over (IN) the (DT) lazy (JJ) dog (NN) . (.)
  - Gold Standard: - The (DT) quick (JJ) brown (JJ) fox (NN) jumps (VBZ) over (IN) the (DT) lazy (JJ) dog (NN) . (.)

Evaluate the performance of the POS tagger by calculating Precision, Recall, F1-score, and Accuracy for the predicted tags with respect to the gold standard.

- C) Explain how the Noisy Channel Model is used in Part-of-Speech (POS) tagging by describing the sequence of steps involved in the process. (2)

- 5) Evaluate the following **Context-Free Grammar in Chomsky Normal Form (CNF)**: (4)

- A) Use the **CKY Algorithm** to parse the sentence:  
**The old man saw the boy with the telescope**

|                                 |                                        |
|---------------------------------|----------------------------------------|
| 1. $S \rightarrow NP VP$ (1.0)  | 7. $PP \rightarrow P NP$ (1.0)         |
| 2. $VP \rightarrow V NP$ (0.5)  | 8. $Det \rightarrow the$ (1.0)         |
| 3. $VP \rightarrow VP PP$ (0.5) | 9. $V \rightarrow saw$ (1.0)           |
| 4. $NP \rightarrow Det N$ (0.3) | 10. $P \rightarrow with$ (1.0)         |
| 5. $NP \rightarrow Adj N$ (0.1) | 11. $Adj \rightarrow old$ (1.0)        |
| 6. $NP \rightarrow NP PP$ (0.4) | 12. $N \rightarrow man$ (0.4)          |
|                                 | 13. $N \rightarrow boy$ (0.3)          |
|                                 | 14. $N \rightarrow telescope$<br>(0.3) |

- Construct the CKY table.**
- Show which grammar rules are used in each cell.
- State whether the sentence belongs to the language of this grammar
- Compute the probability of each parse and state which parse is more likely.

- B) Analyse the sentence how it is structurally ambiguous.  
**"I saw the man with the telescope."** (4)

$S \rightarrow NP VP$   
 $NP \rightarrow Pronoun \mid ART N \mid ART N PP$   
 $VP \rightarrow V NP \mid V NP PP$   
 $PP \rightarrow P NP$   
 $ART \rightarrow the$   
 $N \rightarrow man \mid telescope$   
 $V \rightarrow saw$   
 $P \rightarrow with$   
 $Pronoun \rightarrow I$

- C) Explain ,What is meant by **parsing as search** in Natural Language Processing? (2)

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| Event Name | Subject |
|------------|---------|
|------------|---------|

058-END SEMSTER-ODD-2025 NPT\_4127:Natural Language Processing

| #            | Course Outcome | Description | Marks     |
|--------------|----------------|-------------|-----------|
| 1            | 1              | CO 1        | 14        |
| 2            | 2              | CO 2        | 10        |
| 3            | 3              | CO 3        | 10        |
| 4            | 4              | CO 4        | 12        |
| 5            | 5              | CO 5        | 4         |
| <b>Total</b> |                |             | <b>50</b> |

\* **NA** - CO Not Selected

| Event Name | Subject |
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058-END SEMSTER-ODD-2025 NPT\_4127:Natural Language Processing

| #            | Level       | No of Questions | Marks (%)        |
|--------------|-------------|-----------------|------------------|
| 1            | Remember    | 0               | 0(0%)            |
| 2            | Understand  | 3               | 6(12%)           |
| 3            | Application | 4               | 12(24%)          |
| 4            | Analysis    | 3               | 12(24%)          |
| 5            | Create      | 5               | 20(40%)          |
| 6            | Evaluate    | 0               | 0(0%)            |
| <b>Total</b> |             | <b>15</b>       | <b>50 (100%)</b> |